Experiment 5(b): One's and Two's Complement Using 8085

AIM:

To write and execute an assembly language program to find the one's and two's complement of given 8-bit data using the 8085 microprocessor.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

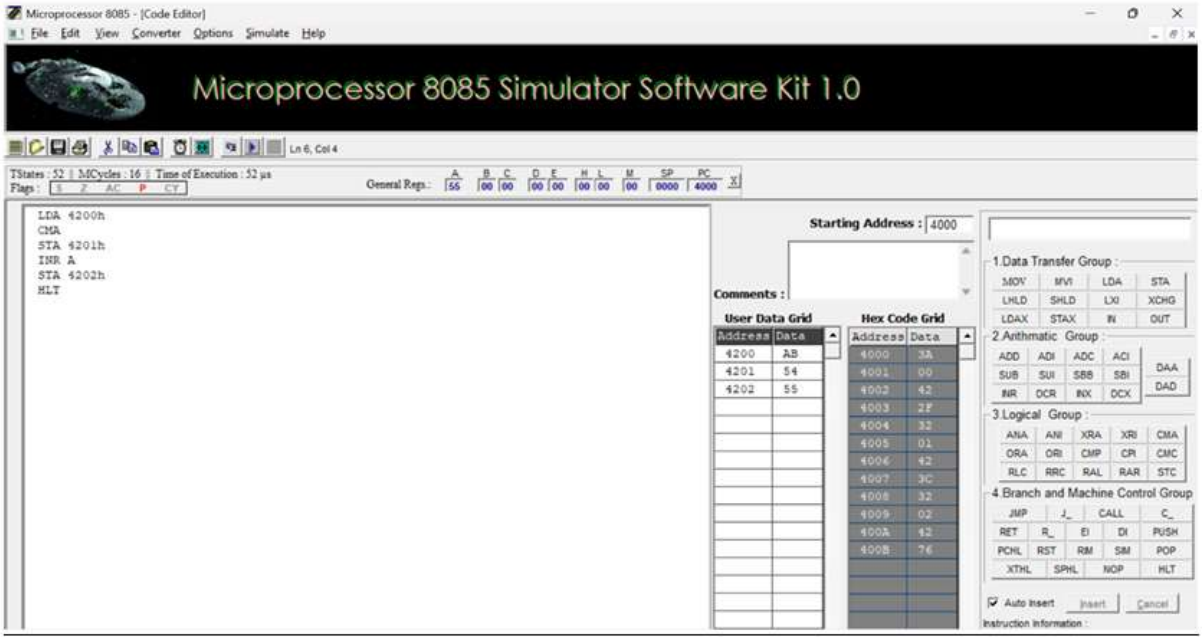
ALGORITHM:

1. Load the data from memory location 4200H into the accumulator.
2. Complement all bits using the CMA instruction.
3. Store the one's complement in memory location 4201H.
4. Increment the accumulator by one.
5. Store the two's complement in memory location 4202H.
6. Stop the program.

PROGRAM:

```
LDA 4200H
CMA
STA 4201H
INR A
STA 4202H
HLT
```

OUTPUT:



RESULT:

Thus, an assembly language program to find the one’s and two’s complement of given 8-bit data was written and executed successfully using the 8085 microprocessor.